

Intelligent Vehicular Emergency Communication System

Team Head & Dept : PRAWIN V S (III ECE B)	
Department	Students
1) III ECE B	1) Prawin V S (23EC084)
2) III ECE B	2) Sanjay Narayanan (23EC101)
3) III ECE B	3) Sarvesh R (23EC105)
4) III ECE B	4) Rithik M (23EC091)
5) III CSE C	5) Sanjey S (23CS152)

Problem Statement

Automatic accident detection and emergency calling systems are currently limited to high-end luxury vehicles.

There is a need for an affordable, hardware-integrated system that not only detects accidents but also automatically initiates voice calls to emergency contacts without human intervention.

Tech Domain: IoT-Driven Automotive Safety & Emergency Response (Sensor Tech)



Novelty

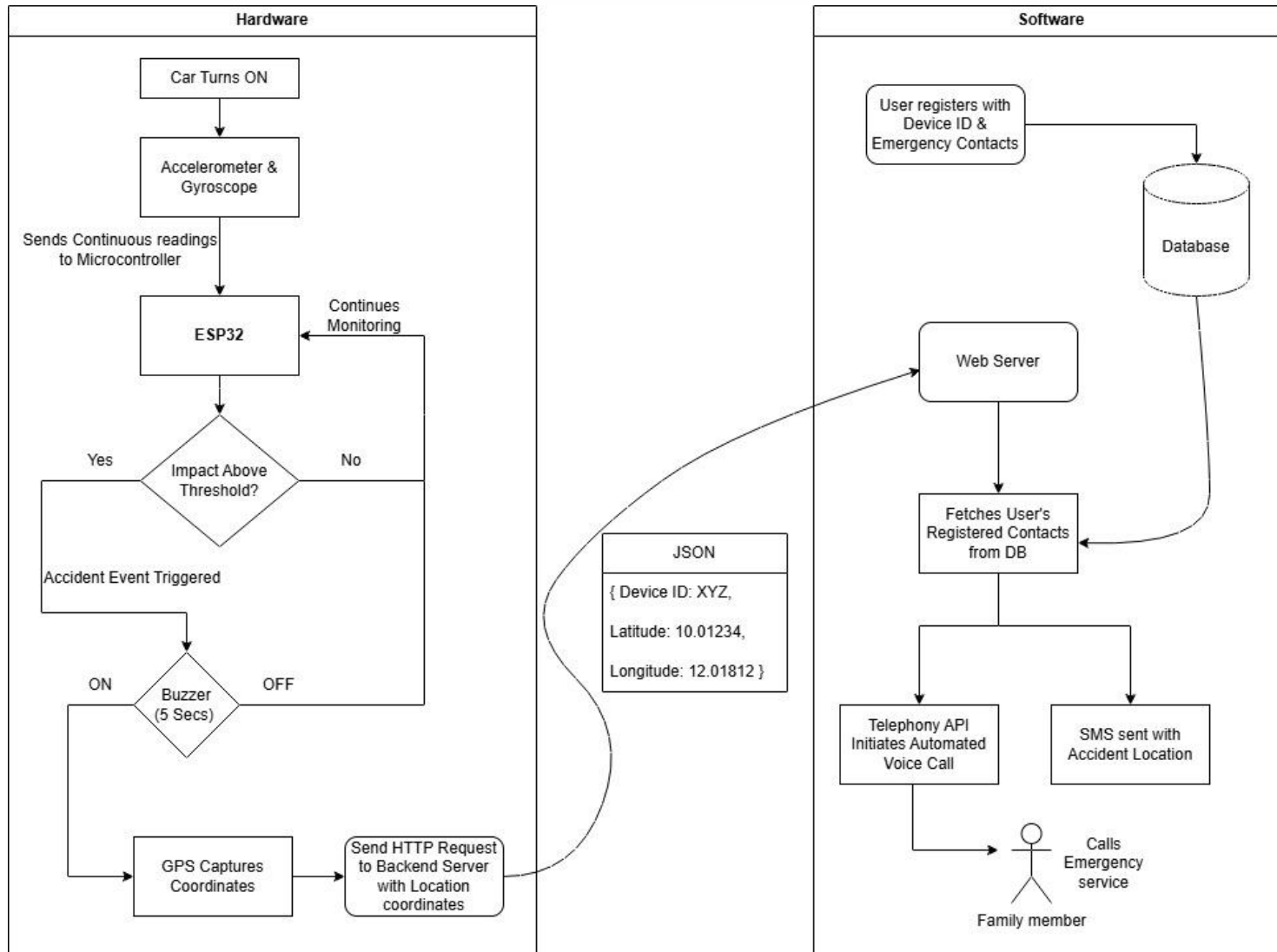
Existing method	Suggested method	Reference
GSM module sends SMS to emergency contacts	Integrates automatic voice calling	Vehicle Accident Emergency Alert System” – IOP Conference Series (2021)
Uses smartphone accelerometer & GPS, Detects impact using g-force threshold	Dedicated ESP32 + sensor hardware, Independent of smartphone battery or placement	Swift Aid: Real-Time Accident Detection and Emergency Alert System” – IRJET (2025)
Uses embedded sensors, Sends data to central monitoring hub/server, Uses machine learning to assess severity	Faster emergency initiation, Lightweight, affordable architecture Designed for easy installation in any vehicle	An IoT-Based Approach to Vehicle Accident Detection and Alerting” – IJCNIS (2025)

Methodology/Flow

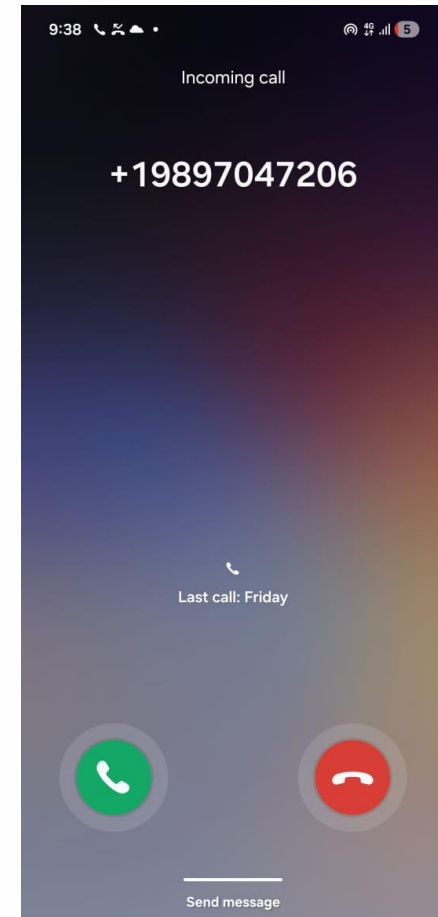


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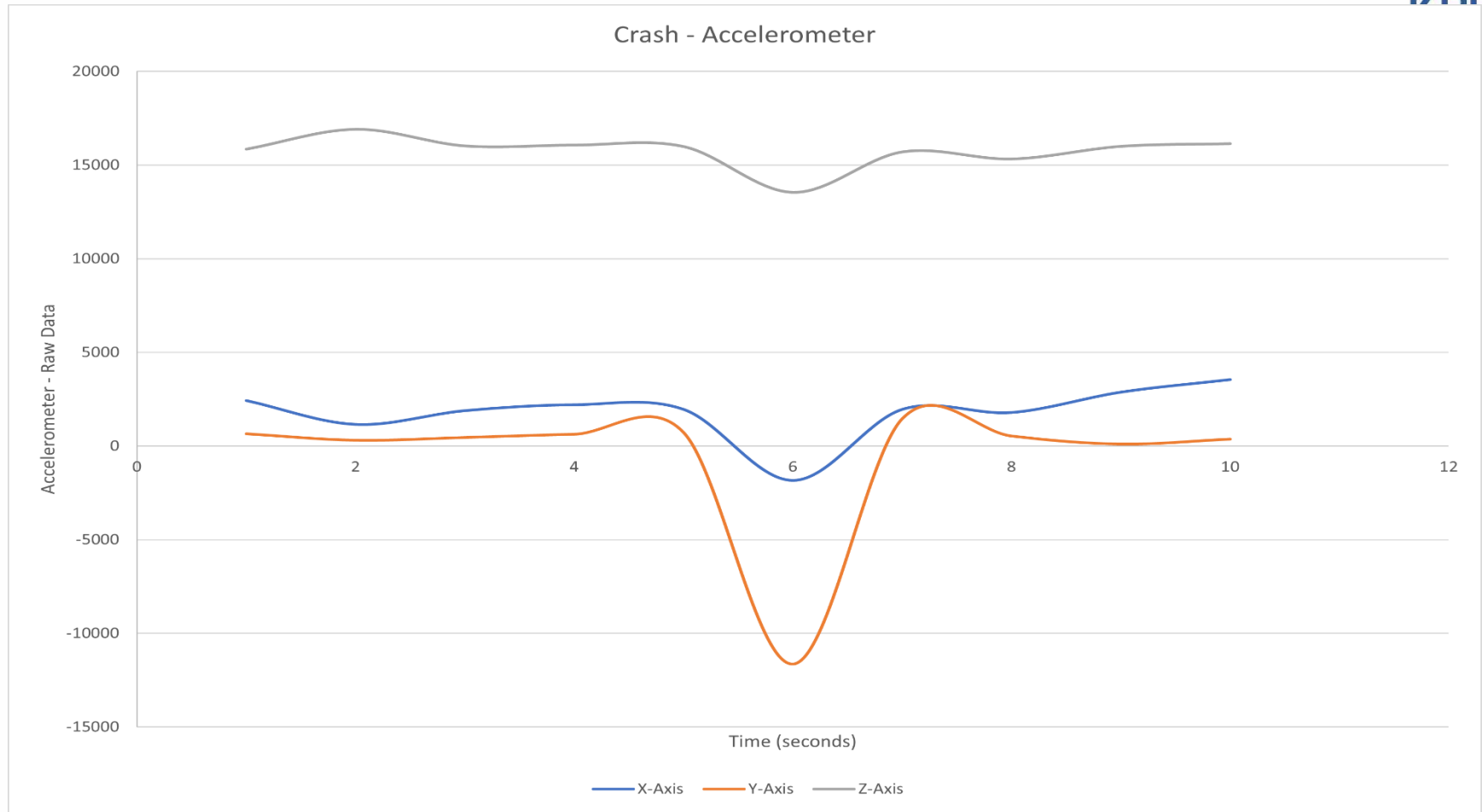
Learn Beyond



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Learn Beyond

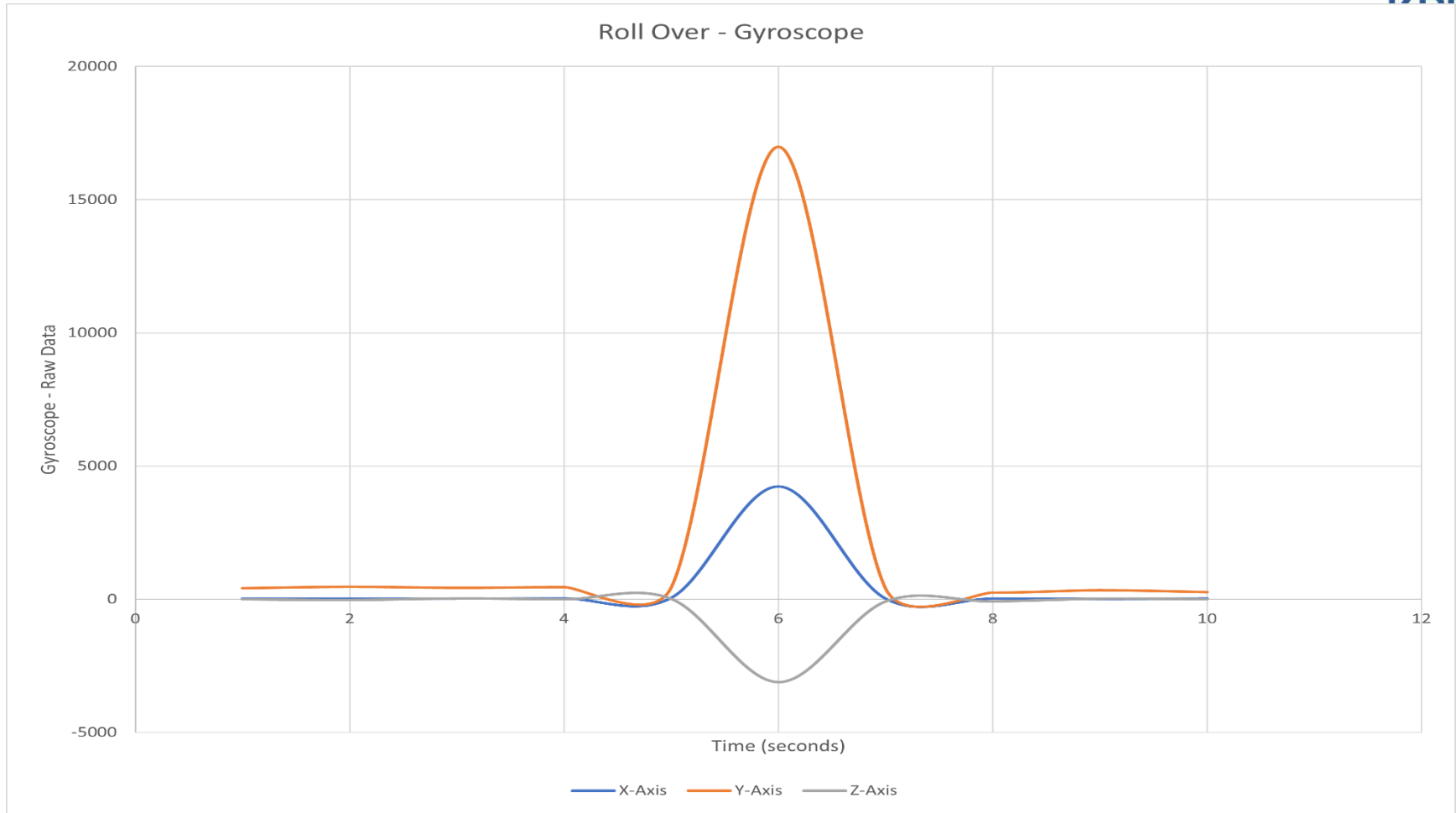


Output Generated



Graphical Representation of Crash Detection through Accelerometer

Output Generated



Graphical Representation of Roll Over Detection through Gyroscope

Output Generated

Reference	Voice Call	SMS with Live Location	Fallback Mechanism (GSM)	Last Tracked Location	Multi Language Support
[1]	No	Yes	No	No	No
[2]	No	Yes	No	No	No
[3]	No	Yes	No	No	No
[4]	No	Yes	No	No	No
[5]	No	No	No	No	No
[6]	No	Yes	No	No	No
[7]	No	Yes	No	No	No
[8]	Yes	Yes	No	No	No
[9]	No	Yes	No	No	No
[10]	No	No	No	No	No
[11]	No	Yes	No	No	No
[12]	No	Yes	No	No	No

Comparing with existing Researches and Publishments
(References)

References

1. “Accident detection using Automotive Smart Black-Box based Monitoring system”, Elsevier Measurement: Sensors 27, 2023.
2. “An IoT-Based Approach to Vehicle Accident Detection and Alerting”, International Journal of Communication Networks and Information Security (IJCNIS), vol. 17, pp. 21-26, Apr. 2025.
3. “Accident Alert and Vehicle Tracking System”, International Journal of Engineering Research and Technology (IJERT), vol. 9, pp. 358-362, Nov. 2020.
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7. “An IoT-Based Automatic Vehicle Accident Detection and Visual Situation Reporting System”, Wiley: Journal of Advanced Transportation. vol. 2024, May 2024.
8. “Android accident detection and alert system”, ICMED: MATEC Web of Conferences, vol. 392, 2024.
9. “Enhancing Vehicle Safety: A Comprehensive Accident Detection and Alert System”, International Journal of Advanced Computer Science and Applications, vol. 14, pp. 28-41, 2023.
10. “IoT-Based Smart Accident Detection and Early Warning System for Emergency Response and Risk Management”, International Journal of Advanced Computer Science and Applications, vol. 16, pp. 660-673, 2025.
11. “Car Accident Detection System And Alert”, International Journal of Advanced Research in Science, Communication and Technology (IJARSCT), vol. 5, pp. 7-11, Apr. 2025.
12. “Vehicle Accident Emergency Alert System”, RJACT: IOP Conf. Series: Materials Science and Engineering, vol. 1012, 2020.

Rate the project Suitability for/patent/research/product/Startup on scale 1(low) -3(High)

Suitability	Scale	Comment
Patent	2	Integration of sensors, GPS, and GSM with hybrid communication shows moderate patent potential
Research	3	Uses sensor data analysis for accident detection with scope for further research
Product	3	Strong real world application for vehicle safety and emergency response
Startup	2	Can evolve into a safety product for fleet monitoring and smart vehicles
Which industry will use this product?	Emergency Services, Transportation Safety	
Are you willing to make the project as a startup?	Yes	